

STRETCH

CEILINGS & WALLS

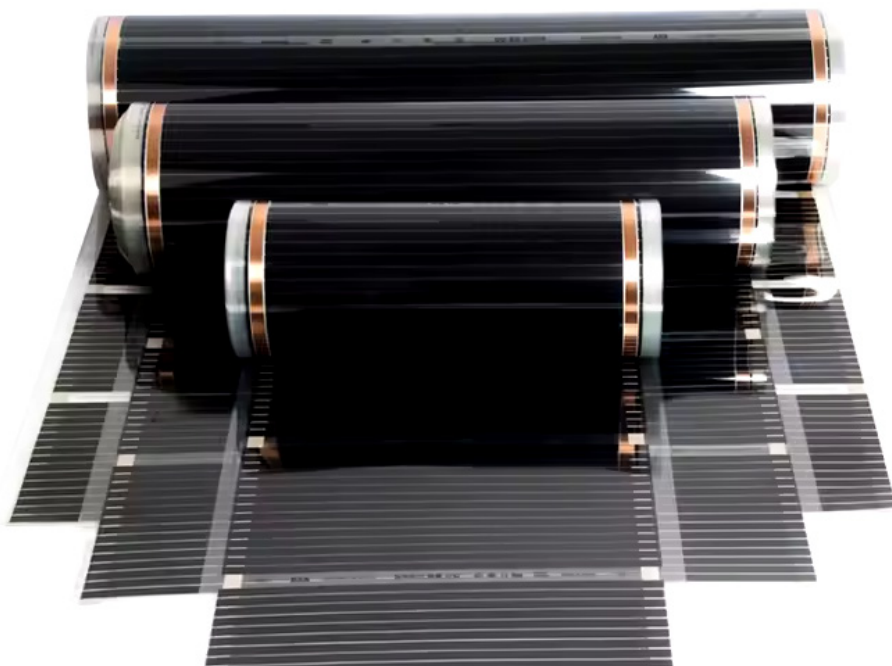
STRETCH IR Heating

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Type	Ceiling/Flooring heating system
Voltage	220V
Power	220 - 280W/sqm
Operating temperature	< 60°C
Length	max 150m
Thickness	0.338mm
Weight	0,50kg/sqm
Warranty	5 Year*
Reference	Net width
S-H-IR-R-33	330mm
S-H-IR-R-60	600mm
S-H-IR-R-80	800mm
S-H-IR-R-100	1000mm

*according to Stretch standard terms of sale

certifié / certified
zertifiziert / заверено
certificado / certificato
ISO 9001 : 2008



Preparation of the floor:

1. Before beginning, please check that all necessary power and provisions are available.
2. Double check with the on-site manager or owner for the exact placement of the heating film.
3. Check with on-site manager or owner to confirm the placement of the thermostat.
4. Check ceiling for moisture of any kind and do not begin installation unless the source of the moisture is identified and removed.
5. Remove the ceiling of all remaining debris and ensure that it is clean.

Installation of the insulation:

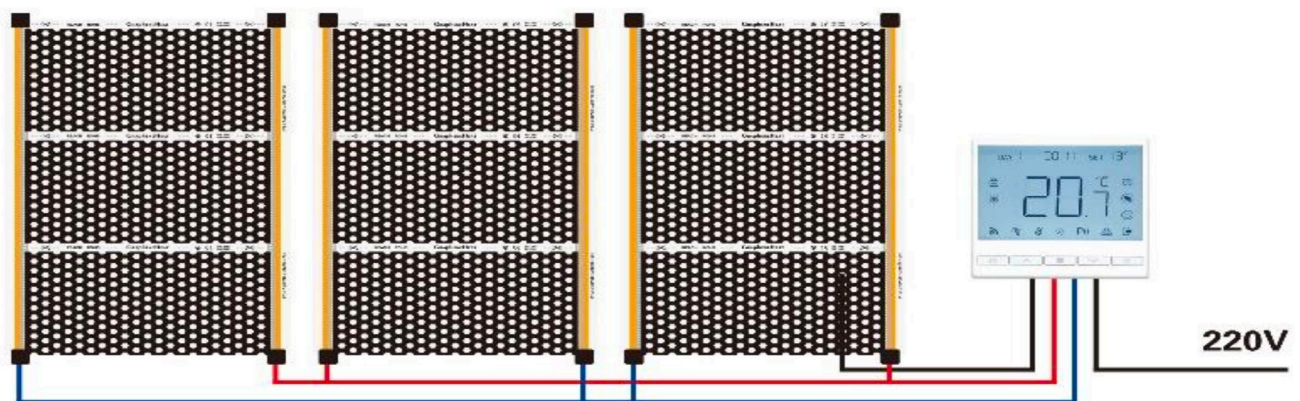
1. The insulating foam should be installed from wall to wall. Take care to note that the insulation should be run in the direction the heating film will be laid out.
2. Use OPP tape to bind each run of the insulation foam. Make sure that the insulation is firmly set against each other.
3. The thickness of the insulating foam may vary the level of the ceiling. To match the siding and the ceiling level, the thickness of the foam should be taken into consideration. 3mm/5mm/8mm insulating foam are available.
4. Make sure that the insulating foam is made for heating film use. Do not use aluminum or Styrofoam.

Installation of the heating film:

1. After preparing the ceiling and installing the insulating foam, make sure the proper width and length of the heating film is measured before initial cut.
2. Please take care to note that the heating film should be placed with the copper bus bar side down.
3. When cutting the heating film, use proper tools and cut along the pre-printed cutting line. Ensure that the copper bus bar will not corrode from external moisture. Ends not terminated should have the protective tape placed on the ends to prevent shorts.
4. Please refer to the chart for power output. This figure shows the power usage in a direct current setup. Do not use series to physically setup heating film.
5. When installing the heating film, make sure the film does not overlap. It is recommended that there remains a space of 1cm between heating films.
6. Heating film should be spaced away from walls at a distance of 10~15cm.
7. Do not install heating film in areas where moisture is likely to appear, such as the front of bathrooms or near kitchen sinks.
8. Do not run electrical wires from the heating film towards moisture prone areas or entrances. They should be placed off to the side and away from traffic and moisture.
9. Do not install heating film where heavy furniture will be set. There are dangers of high heat in those areas.
10. After installation of the insulating foam and the heating film, make sure both are not able to move and fixed in place with OPP tape as necessary.

Electrical connection of Heating film

1. This is one of the most important areas to installing heating film. Mistakes are often made in this step. Please take care to follow instructions.
2. When installing multiple heating films, it is imperative that the films are setup in parallel.
3. Proper gauge electrical wires should be installed in correlation with the wattage being used. Please be aware of the power consumption by calculating the heating film's load and select appropriate gauge wires. Failure to do so may result in improper function of the heating film, thermostat, or both.
4. When installing the heating film in parallel, the electrical wires should be joined and clamped together at the connector. The connector has to clamp ends, one end for the heating film and one end for the wires.
5. Make sure the wires are clamped together first using a clammer. Please be sure to clamp using high pressure to ensure a secure hold.
6. Once the wires are securely clamped, the connectors can be fastened to the heating film. Take care to note that the PET between the copper bus bar and the silver bus bar separate. The connectors fit into that space, clamping on the copper portion of the bar and not the silver bus bar. The back end of the connector will make contact with the silver bus bar. Once positioned correctly, the connector can be clamped. After clamping, make sure that the connector does not move.
7. Please refer to the image for proper terminal connections.



Thermal protection:

1. All terminal ends on the heating film should be protected from shorts and other damages.
2. Protecting terminals is a vital step in assuring proper working order for the heating film system. The bus bar lines are protected by the outer PET film and thus will not short from the outside, but since the bus bar lines separate, protecting the separation points is a must. Both the terminal ends and the isolated ends must be shielded from moisture and or liquids.
3. The connection between the electric heating film and the wire is reliable to prevent false connections. Overloading is strictly prohibited. The wiring end of the electric heating film must be completely sealed with the other end. First, it should be treated with high-temperature resistant waterproof insulation adhesive, and then it should be completely sealed with cloth based tape.
4. The leakage current and static electricity generated by the use of electric heating film must be properly grounded.

Thermostat installation

1. Both electrical wires in the figure section, "A" should be noted that both are connected to the "OUT" on the back of the thermostat.
2. JAE offers thermostats that have sensor function. The use of the sensor provides constant temperature and will automatically turn itself off to remain at desired temperature. The sensor is an optional function and can be switched to manual operation by removing the sensor from the thermostat?
3. When the thermostat is powered without the sensor input, the factory preset will show up on the LED. The thermostat will now be in timer mode. It will turn off and on according to the level set by the user, The thermostat has leveling from 1L to 10H (some thermostats don't have the L r H but will function the same).
4. Once the installation of the thermostat is completed, please test according to step 3 at least 30 minutes to ensure everything is in proper working order;.

Testing and Completion

A. Once installation has been completed, there should be a final inspection made before completion of the ceiling.

1. After electrical connections are made, the thermostat should be checked to see if it is installed correctly.
 - a. If the thermostat has no power, check the circuit breaker and make sure there is power flowing to the thermostat.
 - b. If the circuit breaker switches off, make sure there is no overload on that breaker Please make proper adjustments to safely assign different breakers to each load.
2. If the thermostat is working properly, the ends of the heating film should be checked with a tester for proper wattage load on the film.
3. Properly conducting heating film should be checked for regular and irregular heat, The heating film should be checked to make sure that there are no deformations to the heating from the installation and heating process.
4. Continue to check the breaker and make sure that the voltage load for that breaker remains stable and safe.
5. Once proper working order has been established and there are no problems, final approval should be received by on-site manager or owner before the top floor is completed.

B. Clean up

1. Please remove all debris left over from installation, This includes pieces of insulation, heating film, cardboard and other items including tools. This process must be carried out so that the ceiling may be laid by the next contractor without any damages to the heating film.
2. Please warn the next contractor or persons laying the ceiling to use care when utilizing box cutters, knives and any other cutting tools used to lay the top floor, They should be properly warned to maintain the cleanliness as the ceiling is being installed.
3. A contact number should be left with the on-site manager or owner in case any problems arise.